Convexity Analysis for TSP Solutions

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Introduction

This report analyzes the convexity and similarity patterns of TSP solutions generated using **Greedy Local Search** applied to **random starting solutions**.

The method works as follows:

- Generate multiple random starting solutions
- Apply greedy local search to each random solution
- Analyze the resulting local optima using edge-based and node-based similarity metrics

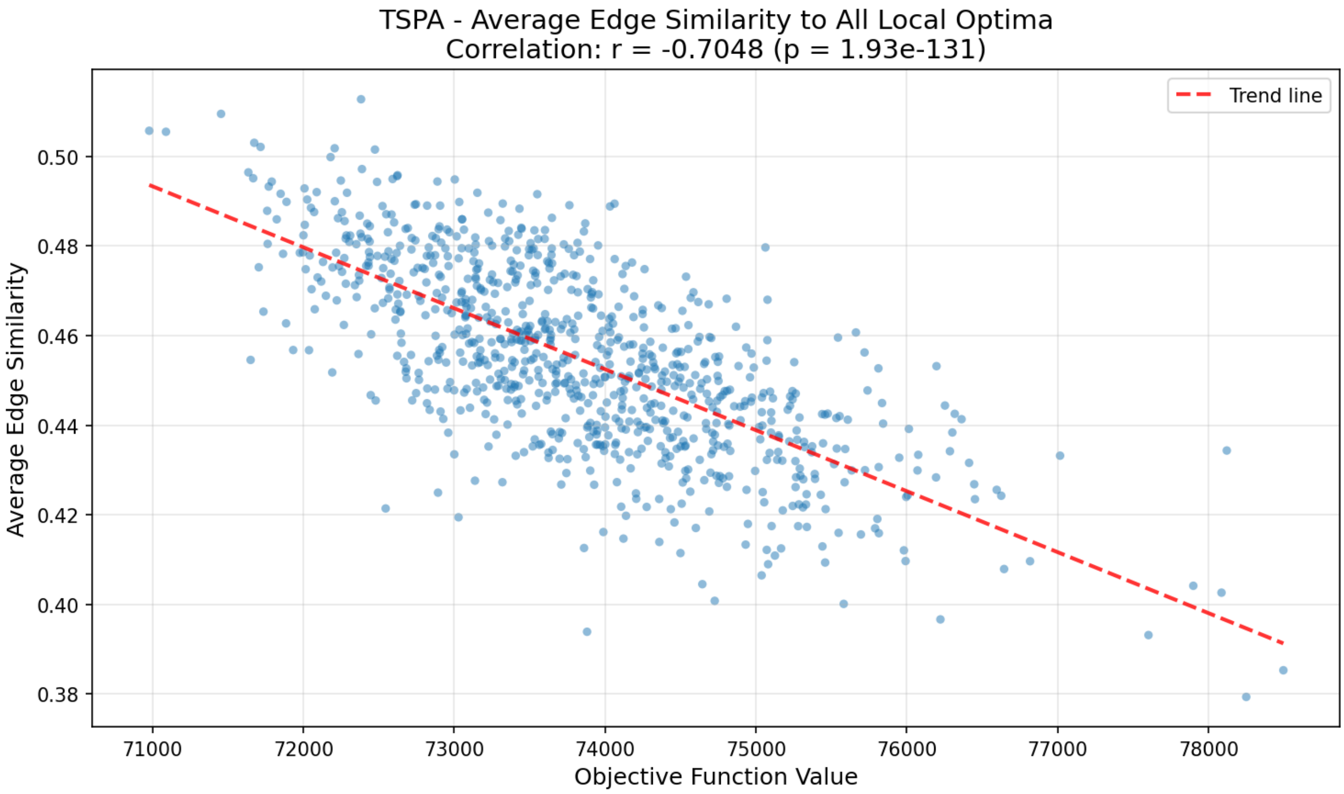
The analysis examines three types of solution comparisons:

1. **Average similarity**
2. **Similarity to the best out of the 1000**
3. **Similarity to the best from other algorithms (Large neighbourhood search)**

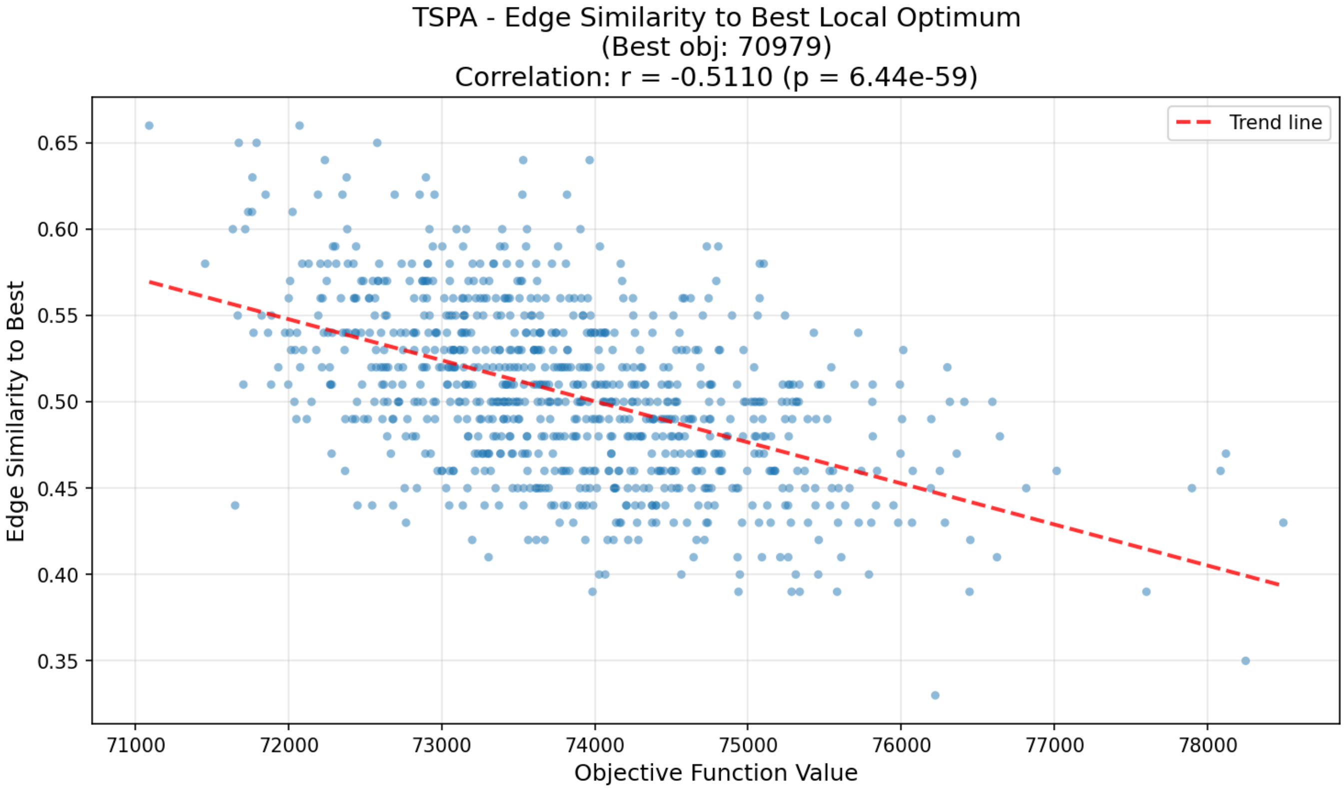
Instance: TSPA

Edge Similarity Analysis

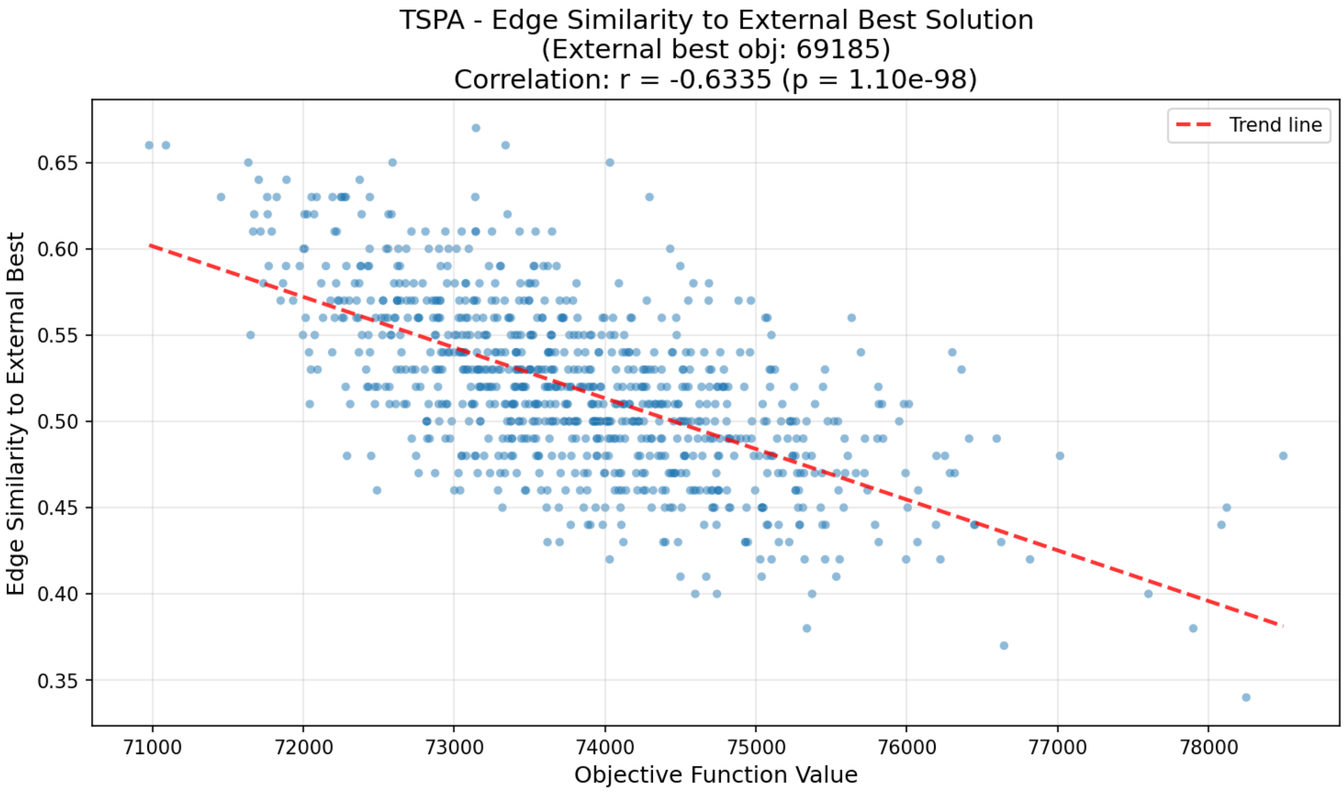
Average - Edge Similarity



Best out of 1000 - Edge Similarity

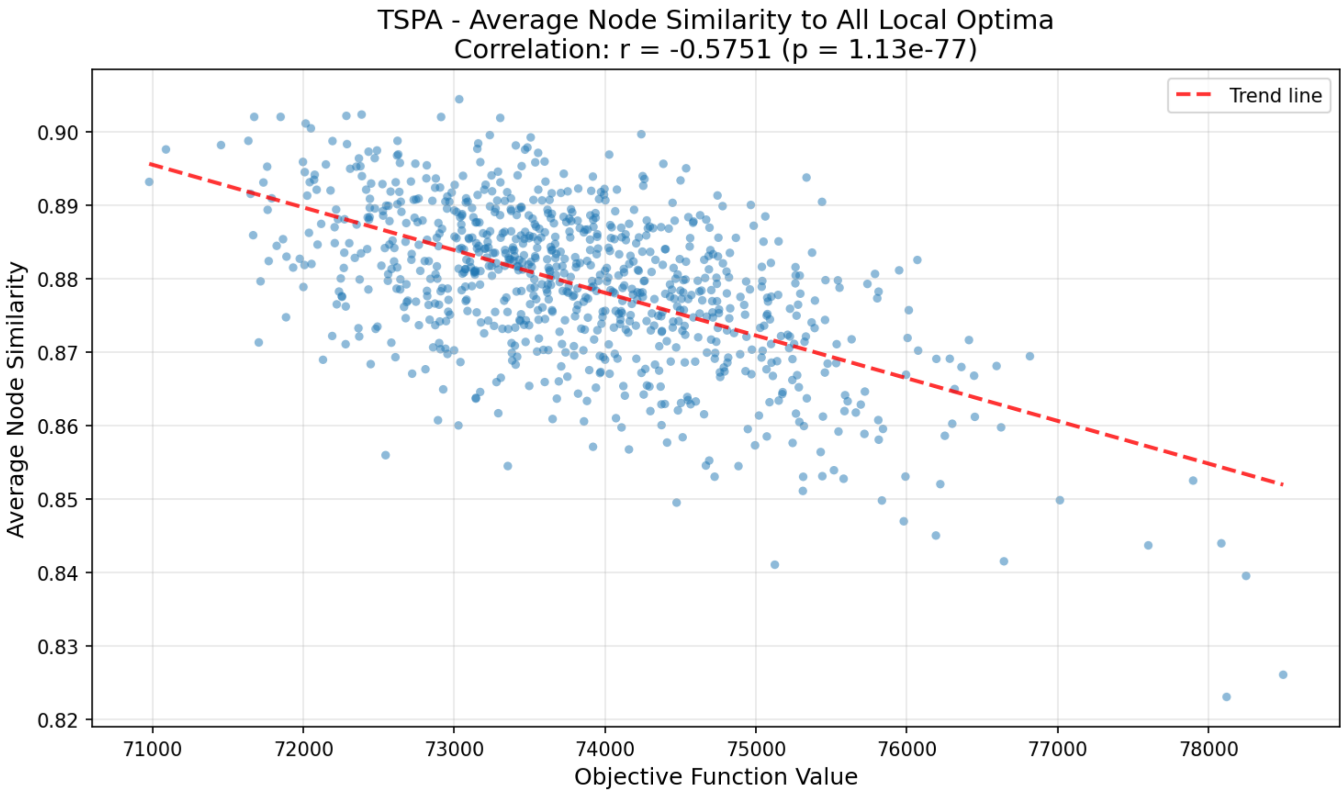


Best External - Edge Similarity

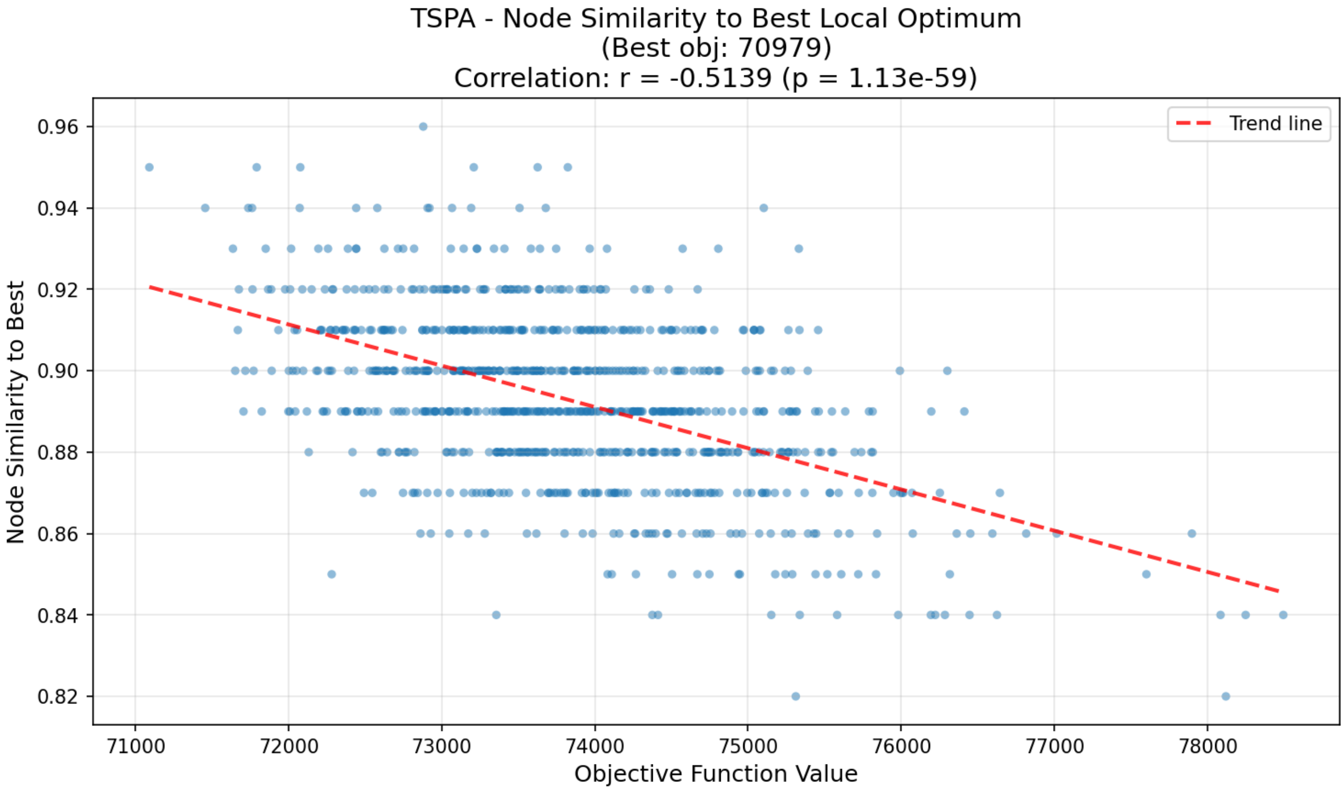


Node Similarity Analysis

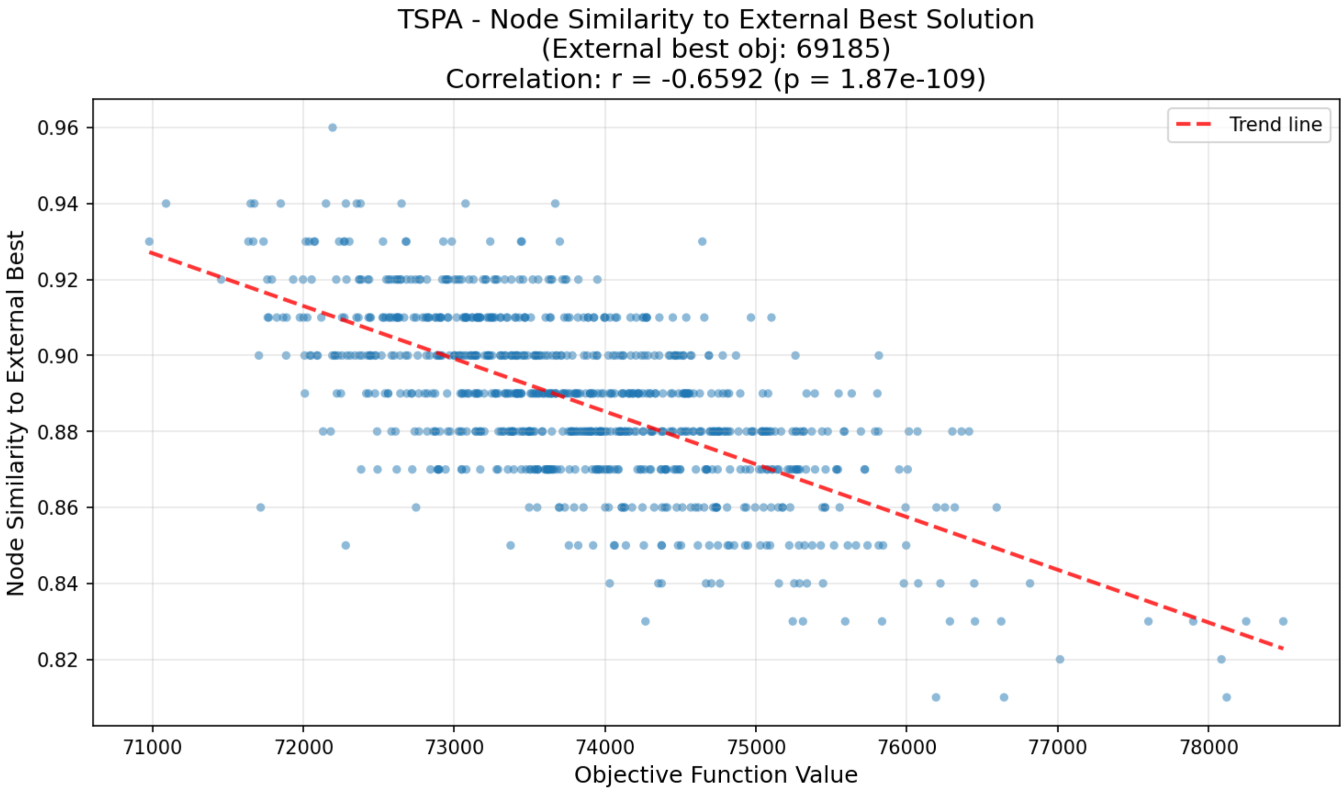
Average - Node Similarity



Best out of 1000 - Node Similarity



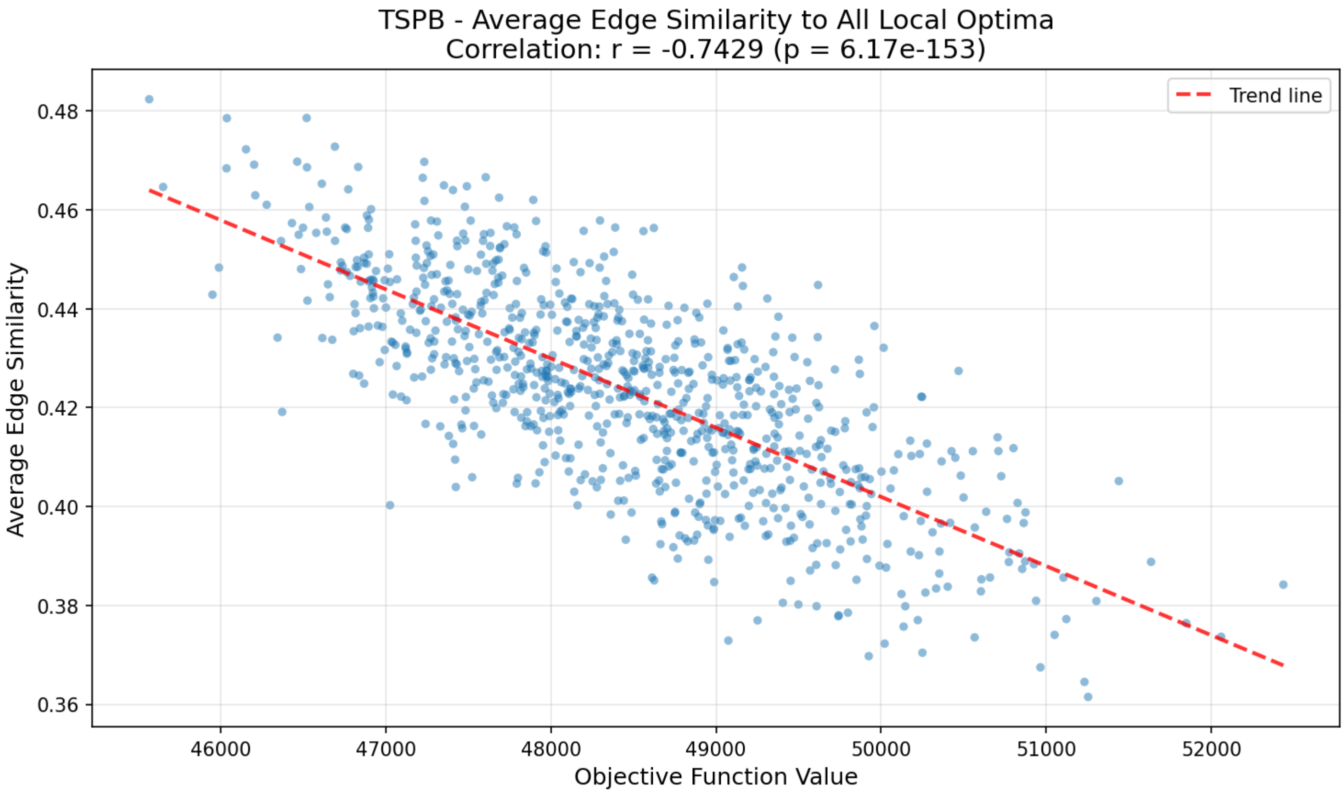
Best External - Node Similarity



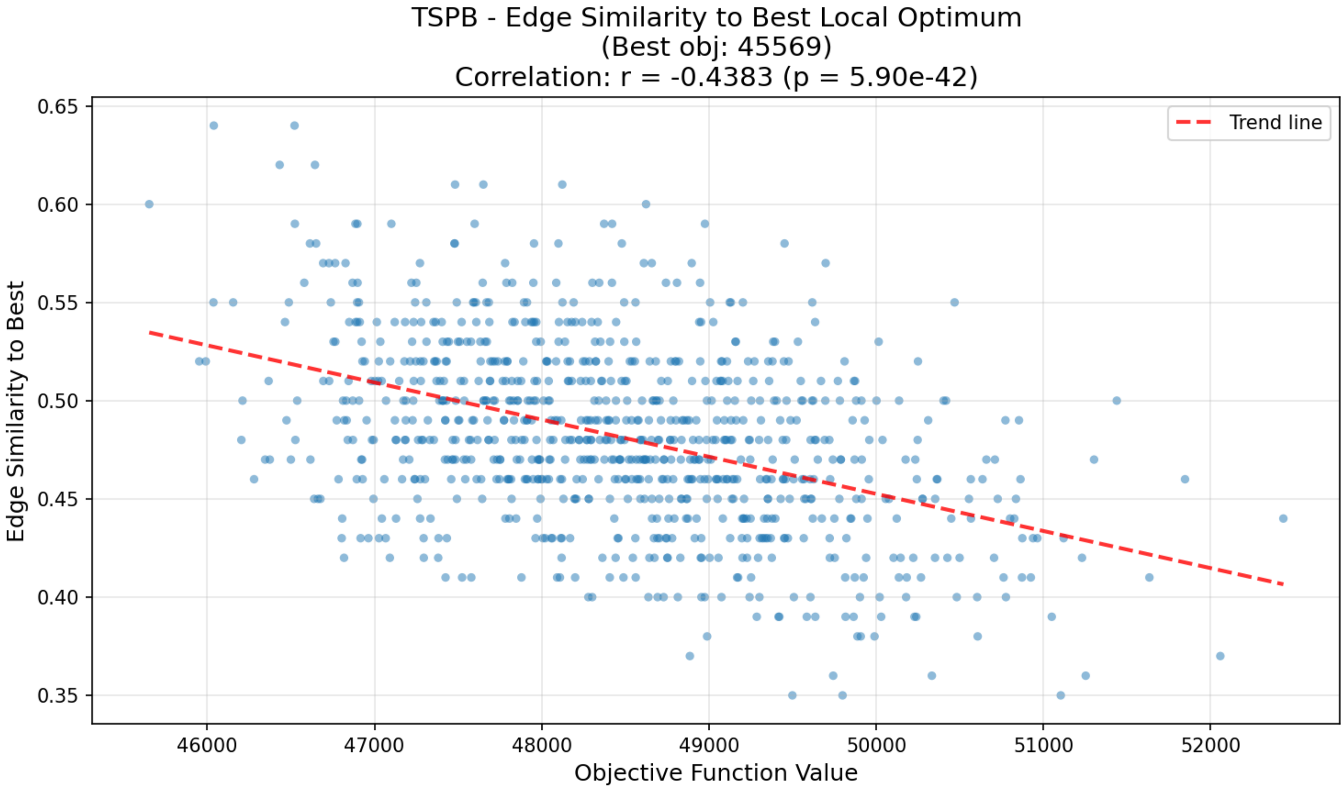
Instance: TSPB

Edge Similarity Analysis

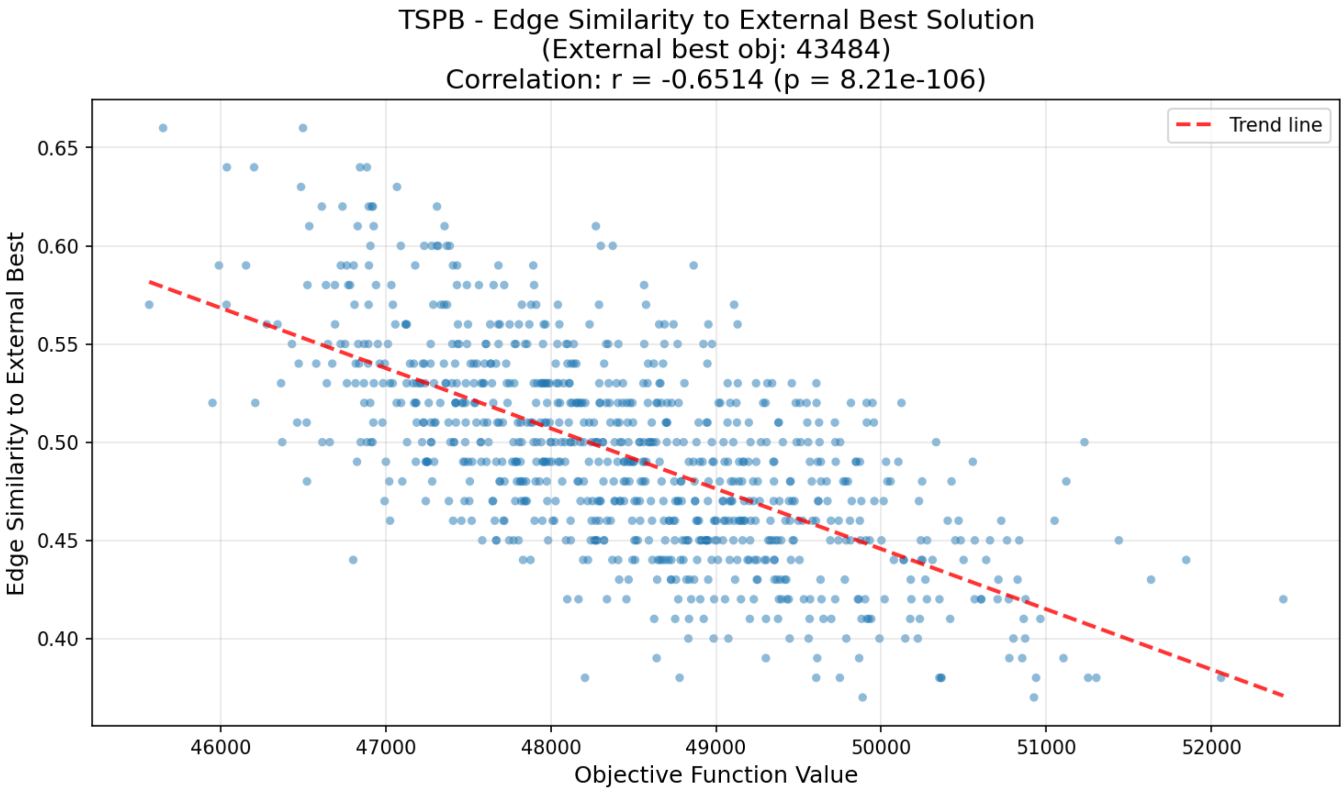
Average - Edge Similarity



Best out of 1000 - Edge Similarity

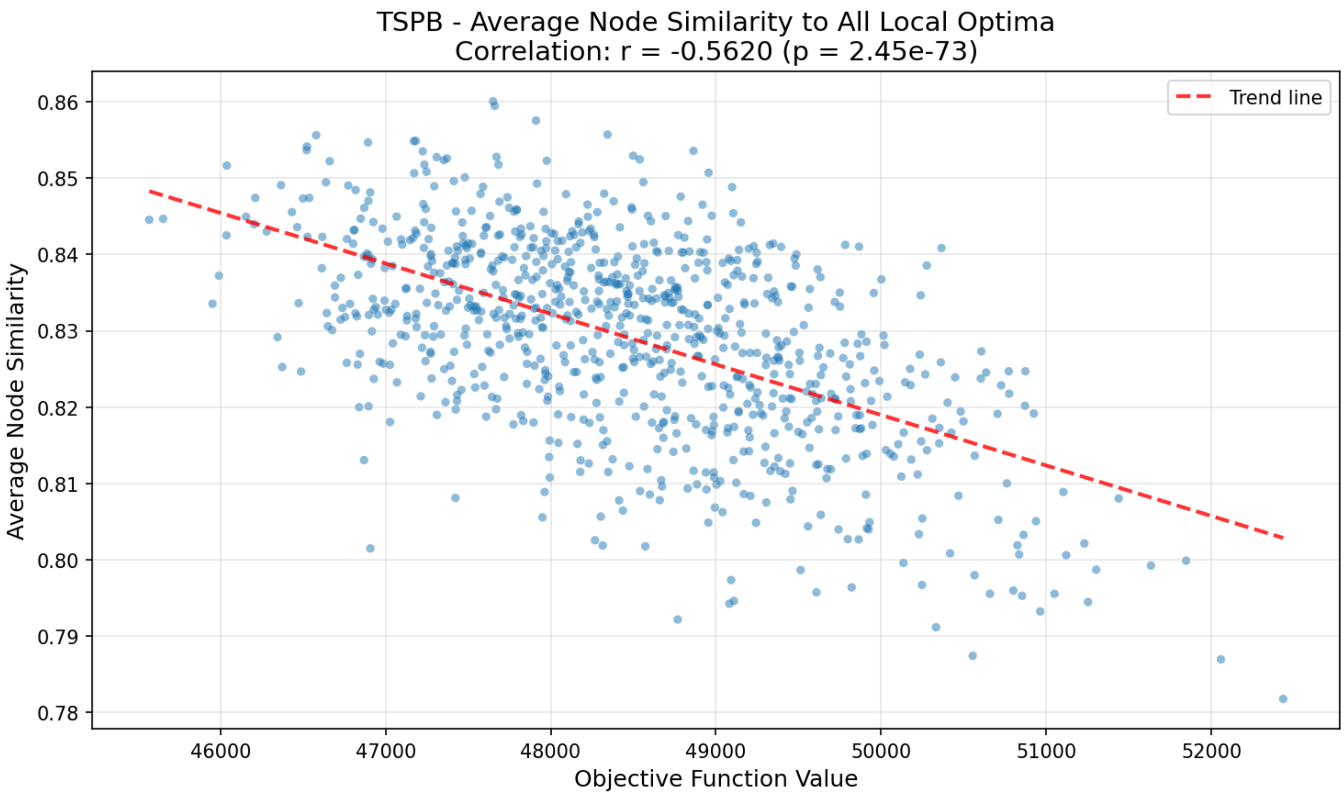


Best External - Edge Similarity

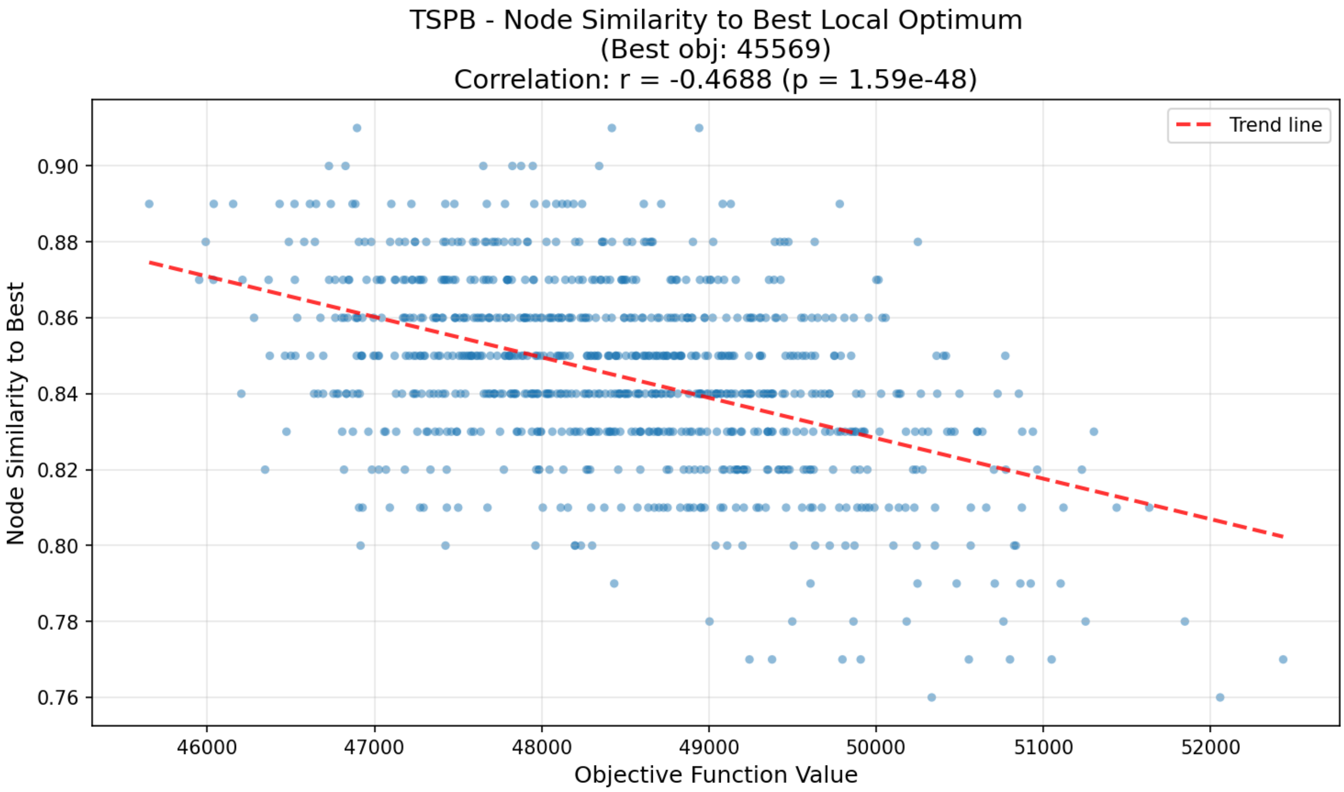


Node Similarity Analysis

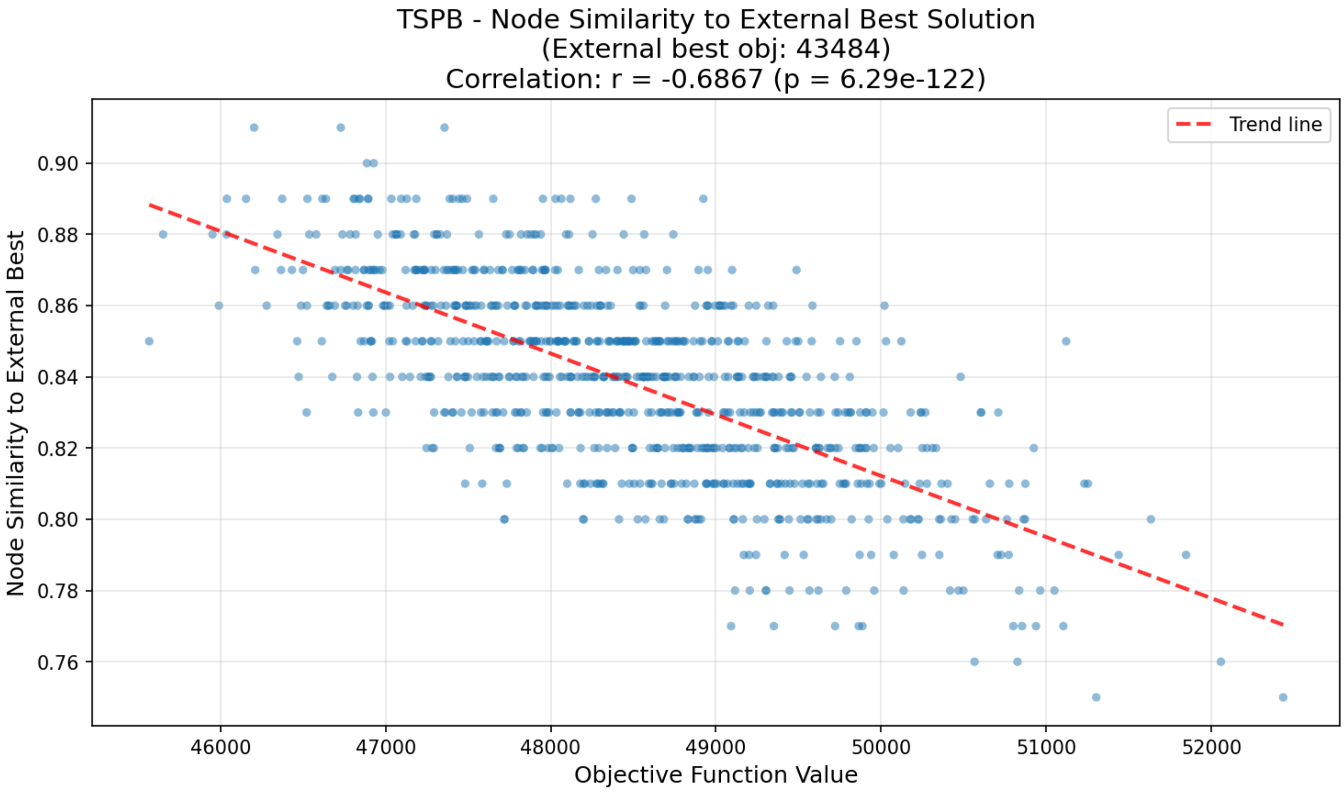
Average - Node Similarity



Best out of 1000 - Node Similarity



Best External - Node Similarity



Summary

The analysis of TSPA and TSPB confirms a "Big Valley" landscape structure, where high-quality local optima are not random but cluster significantly around each other.

Key Findings:

1. Strong Negative Correlation: Across all scenarios, lower objective values (better solutions) correlate strongly with higher similarity.
2. External Validation: Both instances show strong convergence toward the optimum found in the best (so far) algorithm. Notably, TSPB exhibits high correlations for both edge ($r = -0.65$) and node ($r = -0.69$) similarity to the external best.
3. Edge vs. Node: Edge similarity has typically higher correlation to objection function value than node similarity. However, this is not the case for the TSPB global optimum comparison ($r = -0.69$ vs. -0.65).
4. Global vs. Local Gradient: In general solutions are more similar to each other on average than to the best solutions, both generated using the same algorithm as well as the best found so far.

Conclusions

The solution space is resembling convex space, meaning good solutions share structural features.